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Member-5 ROLE - Architecture Design Maker.

TASK -1

1. ML CATEGORIZATION– System that can read transaction names and guess what they are

→ We will implement a transaction categorization engine to automatically classify raw financial transactions into predefined categories such as Food, Transport, Shopping, Utilities, and Income.

In the initial phase, we will use a rule-based keyword matching approach to ensure fast and reliable performance. For example:

- “UBER TRIP” → Transport
- “SWIGGY ORDER” → Food

This approach is simple, efficient, and suitable for real-time processing during early development.

To enhance scalability and intelligence, the system will later be extended using LLM-based few-shot prompting via LangChain, enabling the model to accurately classify previously unseen or ambiguous transaction descriptions.

This hybrid approach ensures both:

- Accuracy (through rule-based logic)
- Flexibility & adaptability (through LLM integration)

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2. FRONTEND / DASHBOARD

→ Title:

“User Interface & Dashboard Design”

The frontend will be developed using Streamlit to provide an interactive and user-friendly dashboard.

Key features include:

- Display of categorized transactions in tabular format
- Visualization of spending patterns using charts
- Category-wise expense breakdown
- Filtering options for better analysis

The dashboard will serve as the primary interface for users to interact with their financial data and gain insights